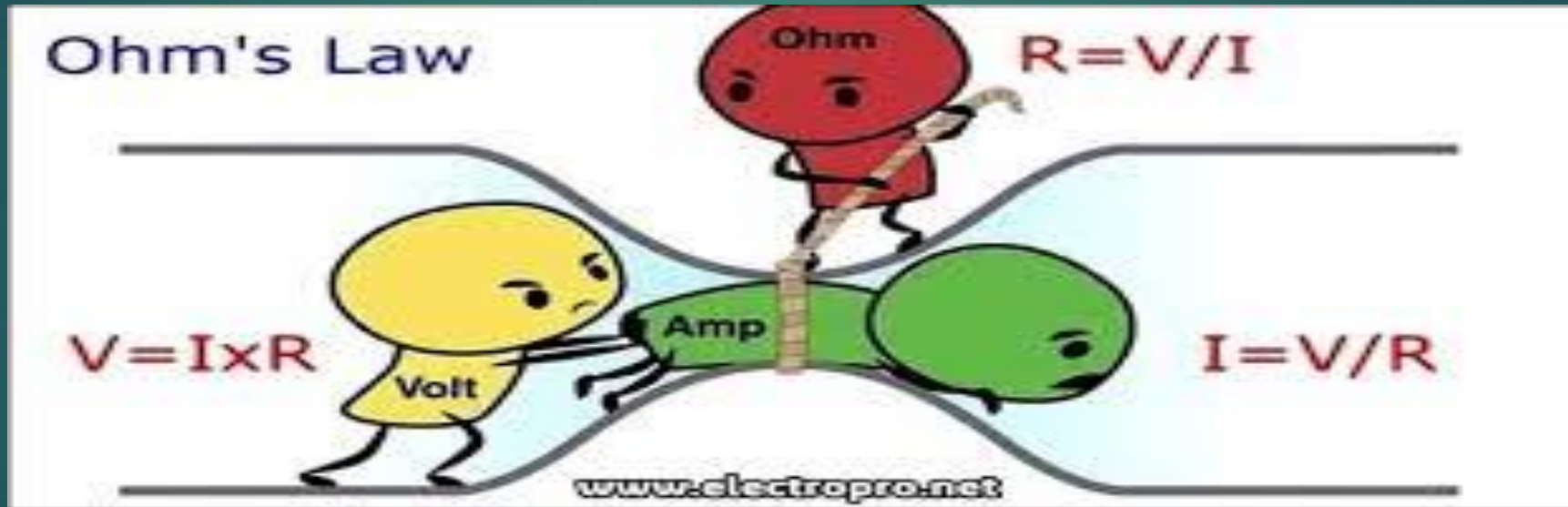


Experiment # (6)

Ohm's Law

قانون أوم



Apparatus

Power Supply



Rheostat



Ammeter



Voltmeter



Banana Wires



Two Resistors



Theory

Ohm's Law states that the current I (in Amp) that flows in a circuit is directly proportional to the voltage V (in volt) across the circuit and inversely proportional to the resistance R (in Ohm Ω) of the circuit:

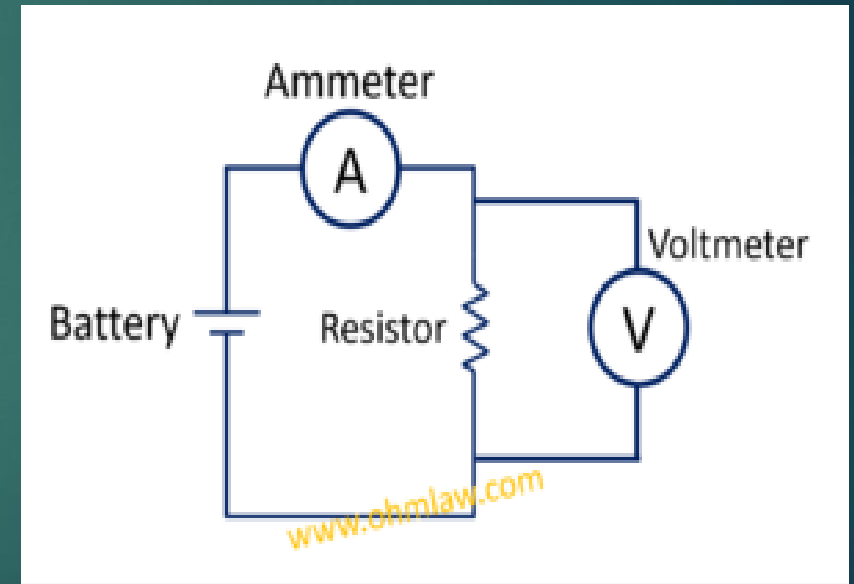
$$V = IR \quad \dots\dots\dots (1)$$

In this experiment, the current flowing through a resistor will be measured as the voltage across the resistor is varied. If the resistors are connected in series then the equivalent resistance is:

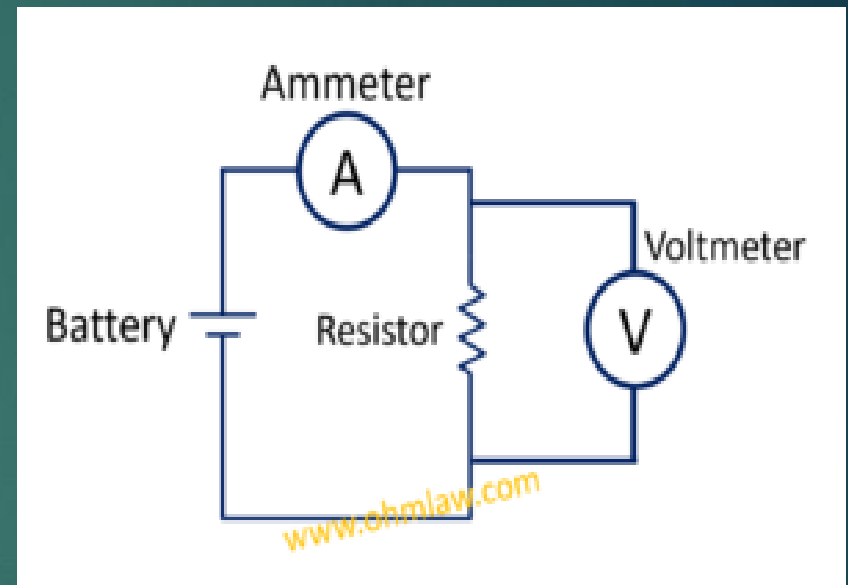
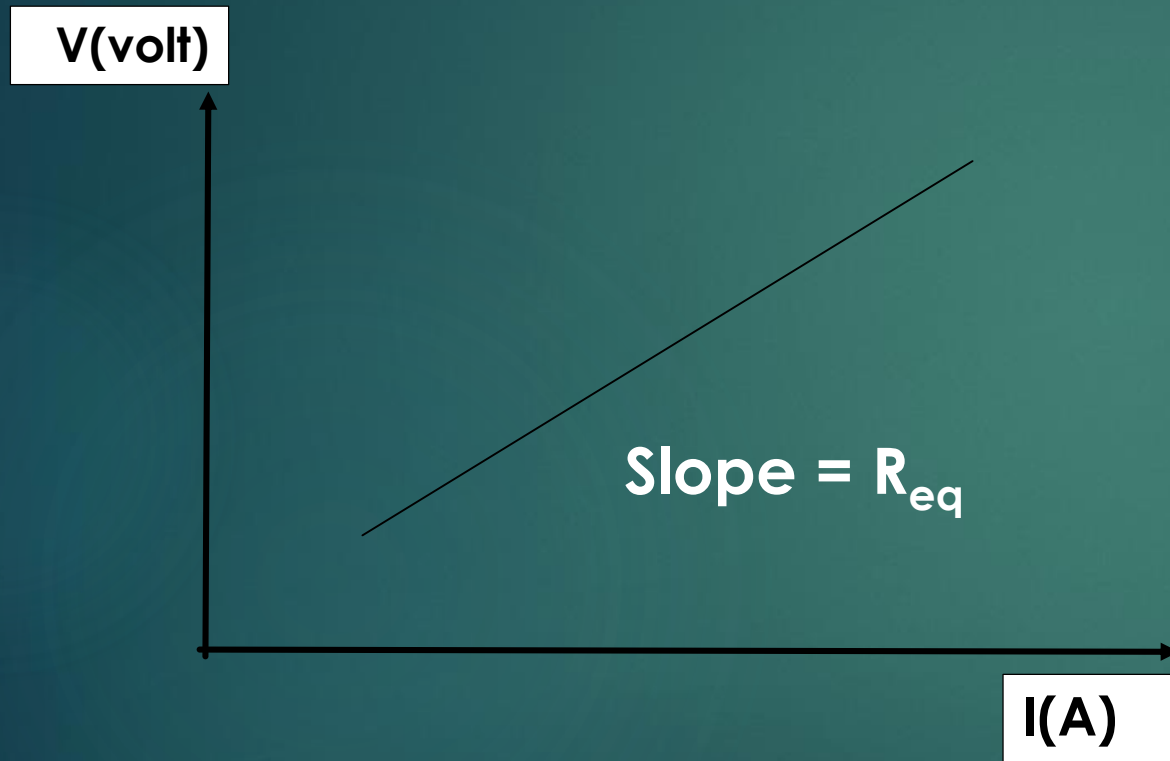
$$R_{eq} = R_1 + R_2 + \dots\dots\dots (2)$$

If the resistors are connected in parallel then the equivalent resistance is:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots\dots\dots (3)$$



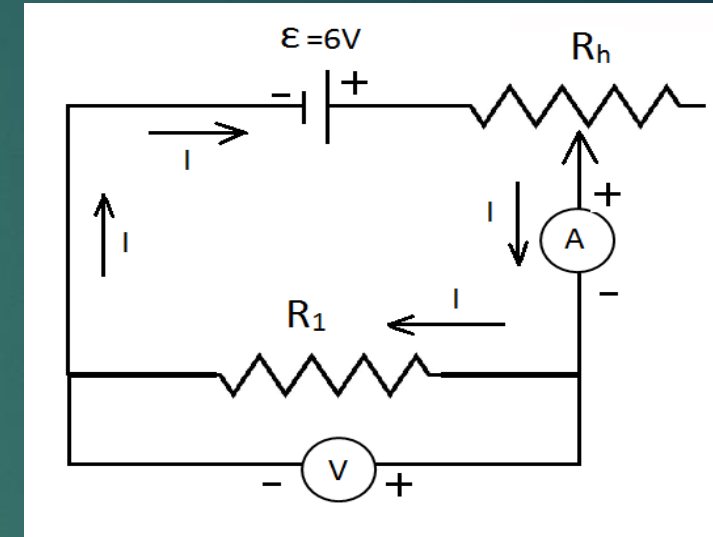
$$V = R_{eq} I$$



Procedure

Part (1): Measuring Resistance

1. Build a simple circuit using R_1 as shown in Fig.1
2. By the rheostat verify that the current and the voltage and measure current and voltage and record them in the table.1.
3. Plot I vs. V in the graph. And apply a linear fit line.
4. From the graph calculate the slope and from the slope calculate the Resistance R_1 .
5. Repeat steps 1-4 for another resistor R_2 .



Figure(1)

Part (2): Resistors in series and parallel

1. Build a simple series circuit using R_1 & R_2 as shown in Fig.2
2. By the rheostat verify that the current and the voltage and measure current and voltage and record them in the table.1.
3. Plot I vs. V in the graph. And apply a linear fit line.
4. From the graph calculate the slope and from the slope calculate the equivalent Resistance $(R_{eq})_{series}$.
5. Repeat steps 1-4 for a parallel connection of the resistors R_1 & R_2 as shown in Fig.3.

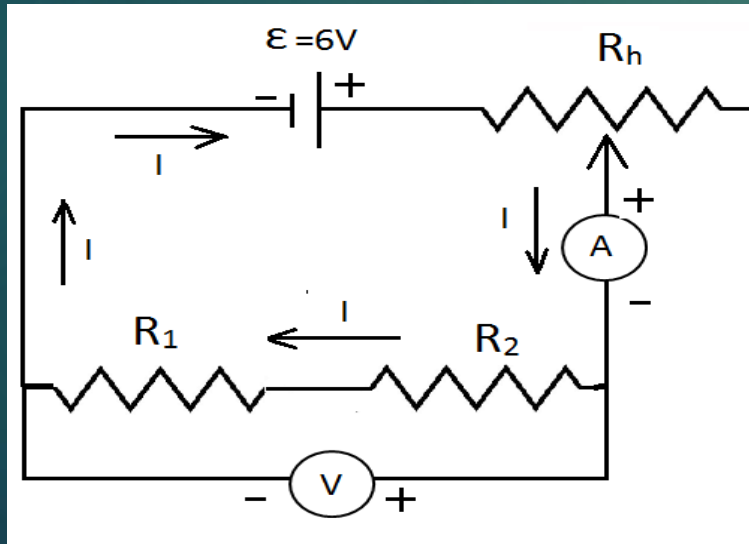


Figure 2: R_1, R_2 in series

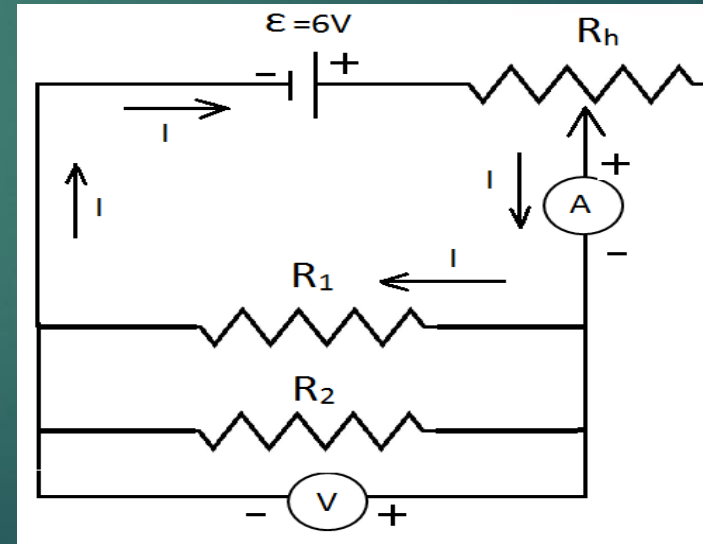


Figure 3: R_1, R_2 in parallel

[illegible]



THANK YOU

